

The Algorithmic Ladder:

RCTs, Artificial Intelligence, and the Future of Human Capital

A Comprehensive Framework for Policy-Relevant Science

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This book provides an exhaustive technical and theoretical guide to the lifecycle of an economic Randomized Control Trial (RCT) applied to Artificial Intelligence in education. Synthesizing Gary Becker's human capital theory, statistical power analysis, and structural econometrics, this framework addresses the "credibility revolution" in empirical economics. Drawing heavily on the Potential Outcomes Framework and real-world examples (such as the Polanco-Jiménez & De Witte AI RCT), the document details protocols for ensuring robustness. It expands the micro-foundations of RCTs to account for historical persistence, institutional political economy, and macroeconomic general equilibrium.

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Part I

The Economics of Digital Learning

Chapter 1

Human Capital in the Age of AI

1.1 The Beckerian Framework: Demand and Supply

To understand the impact of Artificial Intelligence on education, we must ground our analysis in Gary Becker's model of human capital. Individuals and families make reasonable, rational decisions to invest in education: if the discounted present value of the benefits exceeds the costs, they invest.

The Supply Side: AI represents a fundamental technological shock to the supply of education. By reducing the marginal cost of personalized instruction to near zero, AI allows for the mass distribution of elite-level tutoring.

The Demand Side: On the demand side, AI alters the cost-benefit calculus for the student. Adaptive AI reduces the "cognitive friction" and effort cost of learning. However, behavioral constraints (present bias) and social capital deeply influence who actually demands and utilizes these tools.

Chapter 2

Motivation: The Epistemological Foundations of Experimentation

*“Economics is a science as any science can be: the world is our lab, and the many diverse people in it are our subjects ... we approach (our) questions coolly, armed not with passion but with **data**.”*

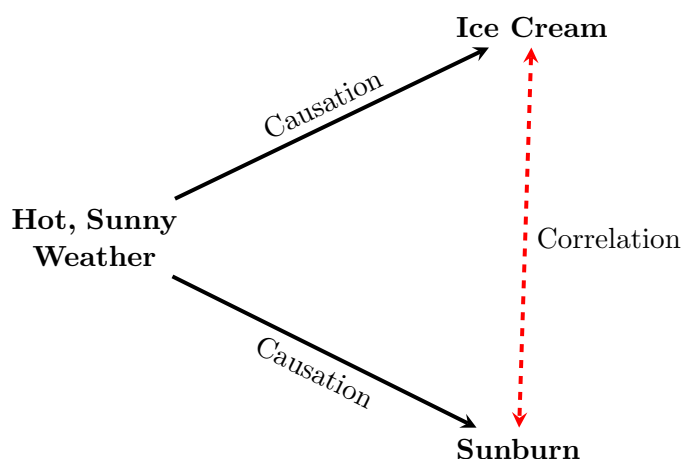
– Angrist & Pischke (2015)

The ascent of the Randomized Control Trial (RCT) to the apex of the empirical hierarchy in economics constitutes a fundamental response to the problem of causal inference. The core objective of an impact evaluation is **causality**—measuring the effectiveness of a program by isolating the true impact from confounding variables.

As Angrist and Pischke note, the metrics craft uses data to get to “**other things equal**” in spite of the obstacles. These obstacles are predominantly **selection bias** and **omitted variable bias (OVB)**.

2.1 Causation vs. Correlation

A classic illustration of the fundamental problem is the relationship between hot summer weather, ice cream sales, and sunburns. While ice cream consumption and sunburns are highly correlated, one does not cause the other. They share a common causal driver.



2.2 The Mechanics of Selection Bias

In the absence of experimentation, economic analysis is perpetually haunted by selection bias. When individuals choose to engage in a program (like downloading an AI tutor), that choice is

rarely random; it is driven by unobservable characteristics (ambition, capability) established by economic theory.

Example: Abraham Wald and Survivorship Bias.

During WWII, Abraham Wald analyzed planes returning from combat with bullet holes. The military wanted to reinforce the areas with the most bullet holes. Wald pointed out that the sample of returning airplanes was *not representative* of all planes that participated in combat. The planes that were hit in critical areas never returned. Analyzing only the survivors introduces massive selection bias.

Part II

The Engine of Causal Inference

Chapter 3

The Potential Outcomes Framework

To formalize this, we use the Potential Outcomes (Rubin Causal Model). Let an evaluation of a social program have:

- **Outcomes** (Y_i): e.g., High school graduation, Financial Literacy Score.
- **Treatment** (D_i): 1 if received AI treatment, 0 otherwise.
- **Controls** (X_i): Age, gender, prior test scores.

For any unit i , there are two potential outcomes:

$$\begin{aligned} Y_{1i} &= \text{Potential outcome for treated} \\ Y_{0i} &= \text{Potential outcome for controls} \end{aligned}$$

The observed outcome Y_i can be written as a **switching regression**:

$$\begin{aligned} Y_i &= D_i Y_{1i} + (1 - D_i) Y_{0i} \\ &= Y_{0i} + D_i (Y_{1i} - Y_{0i}) \end{aligned} \tag{3.1}$$

3.1 The Fundamental Problem of Causal Inference

It is impossible to observe Y_{1i} and Y_{0i} for the same individual simultaneously. Therefore, the Individual Treatment Effect ($Y_{1i} - Y_{0i}$) cannot be calculated. However, we can estimate average parameters:

- **Average Treatment Effect (ATE)**: $E[Y_{1i} - Y_{0i}] = E[Y_{1i}] - E[Y_{0i}]$
- **Average Treatment Effect on the Treated (ATT)**: $E[Y_{1i} - Y_{0i} \mid D_i = 1]$
- **Average Treatment Effect on the Untreated (ATU)**: $E[Y_{1i} - Y_{0i} \mid D_i = 0]$

3.2 Decomposing the Simple Difference

If we run a simple difference in means between treated and untreated groups, taking expectations yields:

$$E[Y_i \mid D_i = 1] - E[Y_i \mid D_i = 0] = \underbrace{E[Y_{1i} - Y_{0i} \mid D_i = 1]}_{\text{ATT}} + \underbrace{\{E[Y_{0i} \mid D_i = 1] - E[Y_{0i} \mid D_i = 0]\}}_{\text{Selection Bias}} \tag{3.2}$$

Selection bias occurs because the counterfactual baseline outcome for the treated group is different from the actual baseline outcome of the control group. RCTs force this bias to zero.

Chapter 4

The Pre-Intervention Infrastructure: Power, Ethics, and Registration

The most common point of failure in economic field experiments occurs before a single data point is collected. The gap between a theoretically valid Potential Outcomes design and a successful field experiment is bridged by rigorous statistical power planning, strict ethical oversight, and transparent trial registration.

4.1 Statistical Architecture: Power Analysis and the MDE

A study with insufficient statistical power has a high probability of yielding a null result even if the true treatment effect is positive and economically meaningful. Conducting an underpowered RCT is not just a waste of financial resources; it is a violation of research ethics.

Statistical power (κ) is the probability of correctly rejecting the null hypothesis when the alternative hypothesis is true. In applied microeconomics, power calculations are framed around the **Minimum Detectable Effect (MDE)**—the smallest true effect size that the study is powered to detect (usually at $\kappa = 0.80$ and $\alpha = 0.05$). For a simple, individually randomized trial, the MDE is approximated by:

$$MDE \approx (t_{1-\alpha/2} + t_{1-\kappa}) \sqrt{\frac{\sigma^2}{P(1-P)N}} \quad (4.1)$$

Where σ^2 is the variance of the outcome variable, P is the proportion of the sample assigned to treatment, and N is the total sample size.

4.1.1 Cluster Randomization and the Design Effect

In many educational AI interventions, it is impossible to randomize at the individual student level due to classroom dynamics or school-wide software licenses. When researchers utilize **Cluster Randomized Trials (CRTs)**, they face a statistical penalty. Observations within the same cluster (e.g., the same school) are correlated, measured by the Intra-Cluster Correlation (ICC, ρ). The MDE must be inflated by the square root of the **Design Effect (DE)**:

$$DE = 1 + (m - 1)\rho \quad (4.2)$$

Where m is the average number of students per cluster. To maximize power and ensure baseline balance, high-quality RCTs employ **Stratified Randomization** (blocking), dividing the sample into subgroups (e.g., high vs. low historical state capacity) before randomizing.

4.2 Ethical Oversight and Institutional Review Boards (IRB)

Economics experiments in the age of AI must adhere to the core principles of the **Belmont Report** (Beneficence, Respect for Persons, and Justice), mediated through rigorous Ethics Committees or Institutional Review Boards (IRB).

- **The Ethics of the Control Group:** A common ethical dilemma in RCTs is withholding a potentially beneficial AI tool from the control group. Researchers solve this via *Phase-In Designs* (the control group receives the AI next semester) or by providing a *Factual Control Baseline* (as seen in Polanco-Jiménez & De Witte, 2026, where the control group still received high-quality factual information, just without the empathetic persona).
- **Data Privacy and Algorithmic Bias:** In digital field experiments, IRBs mandate strict adherence to data protection frameworks (such as GDPR in Europe). Researchers must ensure that the AI algorithms deployed do not inadvertently learn and replicate historical biases against minority or migrant youth.

4.3 Trial Registration and the Pre-Analysis Plan (PAP)

To secure the epistemological boundary between confirmatory and exploratory research, the trial must be registered publicly *before* the intervention begins. Platforms like the **AEA RCT Registry** act as time-stamped ledgers for the social sciences.

The researcher must upload a **Pre-Analysis Plan (PAP)** that details:

1. **Primary and Secondary Outcomes:** The exact construction of the variables (e.g., indices of financial literacy or binary enrollment variables).
2. **Econometric Specifications:** The precise regression equations, including which covariates ($\mathbf{X}_i$) and Fixed Effects will be utilized.
3. **Handling of Attrition and Multiple Testing:** Ex-ante commitments to Lee Bounds or FWER corrections.

By tying their own hands before the data is realized, researchers guarantee that subsequent p-values retain their classical interpretation, immunizing the study against accusations of data-dredging or HARKing (Hypothesizing After Results are Known). This rigid pre-registration lays the groundwork for the dual-track reporting strategies required when unexpected field realities inevitably arise.

Chapter 5

Case Study: Evaluating Applied AI

5.1 The Tailoring Premium (Polanco-Jiménez & De Witte, 2026)

To ground these concepts in modern Applied Microeconomics, consider the RCT conducted by Polanco-Jiménez and De Witte (2026), titled “*The Tailoring Premium: AI Design, Behavioral Engagement, and Human Capital Formation.*”

Policymakers face a “build vs. buy” dilemma: Should they permit off-the-shelf, general-purpose AI models (cheap) or invest in deeply curriculum-tailored AI tools (expensive)? To causally answer this, the authors ran an RCT with 2,440 secondary students in Belgium evaluating financial literacy.

The randomization assigned students to one of three arms:

1. **T0 (Control):** Traditional online instruction (Status Quo).
2. **T1 (Generic AI):** Condensed instruction supplemented by a general-purpose AI chatbot (mimicking off-the-shelf tools like ChatGPT/Gemini).
3. **T2 (Tailored AI):** An adaptive chatbot explicitly engineered for the specific Flemish curriculum and pedagogical structure.

Why this design is crucial: If students were simply allowed to pick their AI tool, the most tech-savvy students might select T2, creating massive selection bias. By assigning access randomly via an algorithm upon login, *no observable or unobservable characteristic affected program assignment.*

5.2 Correcting Information Biases for Minority Students (Polanco & De Witte, 2026)

While scaling AI can reduce the marginal cost of instruction, evaluating its impact requires us to look closely at marginalized populations. In Flanders, Belgium, students with a migration background face a well-documented “migrant penalty”—they are disproportionately channeled into vocational tracks and are less likely to transition to higher education, even after conditioning on academic performance.

This suboptimal enrollment is a classic manifestation of frictions in the Human Capital Model (HCM). Because these youth lack host-country social capital, they suffer from severe information asymmetries regarding financial aid, institutional quality, and the labor-market returns to education. Consequently, they *overestimate costs* and *underestimate benefits*, flipping the Beckerian inequality and leading to under-enrollment.

To test whether AI can correct these biases, Polanco-Jiménez and De Witte (2026) conducted a multi-armed RCT testing not just *if* information matters, but *how* it should be delivered.

5.2.1 Experimental Design and the AI Personas

Drawing on the human advising literature—which suggests that counselor empathy and role-modeling can close achievement gaps—the researchers randomized students into interacting with one of four AI personas:

1. **Control (Factual Bot):** Provides structured, neutral information to reduce the information constraint.
2. **Friendly Guide:** Adds conversational warmth to target the engagement constraint.
3. **Empathetic Mentor:** Provides psychological scaffolding to target the self-efficacy constraint.
4. **Inspiring Storyteller:** Uses relatable narratives to target the identity constraint (social proof).

5.2.2 Average Effects vs. Essential Heterogeneity

The results of the trial highlight the danger of relying solely on the Average Treatment Effect (ATE). On average, the researchers found **no significant positive impact** of the warmer personas compared to the factual Control. Simply “humanizing” an AI does not automatically increase educational aspirations; high-quality, factual information is a powerful baseline.

However, moving beyond the ATE reveals critical behavioral heterogeneity, validating a structural approach to RCTs.

- **Engagement vs. Persuasion:** While the personas did not persuade the average student to enroll, the *Friendly Guide* successfully reduced study attrition by 13.5 percentage points. It solved an engagement friction, even if it didn’t shift the final human capital calculus.
- **The Danger of Artificial Empathy for the Present-Biased:** The researchers analyzed treatment effects conditionally based on a student’s time preference (patience). For “Impatient” (present-biased) students, interacting with the *Empathetic Mentor* significantly **reduced** their enrollment probability by 14.9 percentage points ($p < 0.01$) compared to the factual Control.

5.2.3 Mechanism and Policy Implications

Why did an empathetic AI harm the aspirations of present-biased students? Mediation analysis revealed that the conversational warmth of the *Friendly* and *Mentor* personas significantly decreased the **perceived clarity** of the information. For myopic students who already struggle with cognitive load and future-orientation, “conversational noise” obscures the hard facts needed to make a high-stakes decision.

The policy implication is profound: “Warmer” is not always better. The design of educational AI cannot be a one-size-fits-all approach. To optimize human capital formation, AI systems must be dynamically adaptive—assessing a student’s behavioral profile (like time preference) and modulating between factual clarity and psychological scaffolding accordingly.

Chapter 6

Core Experimental Assumptions and Verification

For an RCT to successfully identify causal effects, two baseline assumptions must hold:

1. **SUTVA (Stable Unit Treatment Value Assumption):** The treatment status of individual i does not affect the treatment status or outcomes of individual j ($i \neq j$). There is no interference or spillover. It also implies treatments are homogeneous ("no variations in treatment").
2. **Ignorability (Independence):** Potential outcomes are independent of treatment assignment: $(Y_1, Y_0) \perp D$. This is achieved directly through the random allocation of the program.

6.1 Verifying Randomization (Balance Tests)

If a treatment is assigned randomly, there should be no correlation between baseline characteristics (ex-ante observables) and treatment status. We verify this via:

- **Covariate Balancing:** Regress each individual characteristic X_i against the treatment D_i : $X_i = \alpha + \rho D_i + v_i$. The coefficient ρ should be statistically indistinguishable from zero.
- **Joint Significance:** Regress treatment against all observables simultaneously: $D_i = \alpha + \rho_1 X_{1,i} + \dots + \rho_k X_{k,i} + v_i$. Perform an F-test where $H_0 : \rho_1 = \dots = \rho_k = 0$.

Part III

Advanced Econometrics and Structural Behavior

Chapter 7

Imperfect Compliance and Econometric Estimation

Even with perfect randomization, fieldwork rarely perfectly translates assignment into actual treatment. Participants might refuse treatment, or control group members might find a way to access the program. This introduces **Imperfect Compliance**.

Let Z be the random assignment (the lottery offer) and D be the actual treatment received. This creates four sub-populations:

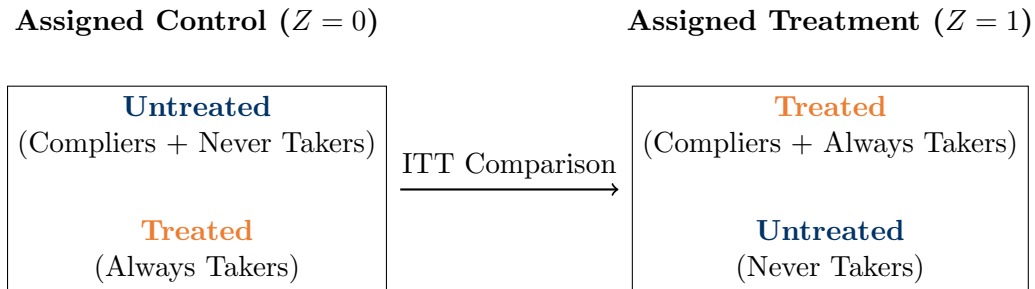
Table 7.1: The Four Types of Compliance

Group Name	Behavior	Assigned (Z)	Treated (D)
Compliers (ϕ_C)	Do exactly what they are assigned to do.	$0 \rightarrow$ $1 \rightarrow$	0 1
Never Takers (ϕ_N)	Untreated despite the offer.	$0 \rightarrow$ $1 \rightarrow$	0 0
Always Takers (ϕ_A)	Treated anyway, regardless of assignment.	$0 \rightarrow$ $1 \rightarrow$	1 1
Defiers	Do the opposite of assignment.	$0 \rightarrow$ $1 \rightarrow$	1 0

Notes: Under the monotonicity assumption, the defier group is assumed to be empty.

7.1 Visualizing Compliance

When compliance is imperfect (Compliance rate $\phi_C < 100\%$), the assigned Treatment Group contains both *Treated* (Compliers + Always Takers) and *Untreated* (Never Takers). The Control Group contains *Untreated* (Compliers + Never Takers) and *Treated* (Always Takers).



7.2 Intent-to-Treat (ITT) and Local Average Treatment Effect (LATE)

Because we only know assignment Z for sure, comparing the entire assigned Treatment group to the assigned Control group gives us the **Intent-to-Treat (ITT)** estimator:

$$Y_i = \alpha + \beta_{ITT}Z_i + u_i \quad (7.1)$$

ITT is highly policy-relevant because governments can only *offer* programs, not force compliance. Mathematically, the ITT is simply the LATE scaled by the compliance rate (ϕ_C). Therefore, LATE can be estimated using Two-Stage Least Squares (Instrumental Variables):

$$LATE = \frac{E[Y \mid Z = 1] - E[Y \mid Z = 0]}{E[D \mid Z = 1] - E[D \mid Z = 0]} = \frac{ITT}{\phi_C} \quad (7.2)$$

Chapter 8

Beyond the Black Box: Structural Heterogeneity

8.1 Essential Heterogeneity and MTE

While LATE identifies the effect for compliers, it masks *essential heterogeneity* (Heckman). Families sort into AI usage based on unobserved expected gains. The "marginal" student may benefit far less than the "average" student. By varying the propensity to participate through experimental incentives, we trace the **Marginal Treatment Effect (MTE)**. This structural approach allows policymakers to forecast the effects of universal AI adoption versus targeted subsidies.

Chapter 9

Navigating Field Realities: Spillovers and Attrition

The theoretical elegance of the Potential Outcomes framework frequently collides with the complex realities of implementing field experiments. When deploying AI tools in real-world educational settings, researchers must manage threats to both internal validity and the Stable Unit Treatment Value Assumption (SUTVA).

9.1 SUTVA Violations and Digital Spillovers

In the age of digital learning, information is highly porous. If a student in the treatment group receives an exceptionally helpful AI chatbot, they may share the output, or the device itself, with a peer in the control group. This creates a **spillover effect**, violating SUTVA. To correct for this, modern RCTs employ *Cluster Randomization* (e.g., randomizing at the school or classroom level rather than the student level) or *Saturation Designs*, which randomize the proportion of treated individuals within a cluster to explicitly measure the indirect network effects.

9.2 Differential Attrition and Lee Bounds

Attrition is the silent killer of RCTs. If students in the factual Control group find the intervention boring and drop out of the study at higher rates than those in an engaging AI Treatment group (as seen with the *Friendly Guide* persona), the remaining sample is no longer balanced. The treatment group will contain low-motivation students, while the control group will only contain high-motivation survivors.

To lead an analysis with severe attrition, researchers must employ **Lee Bounds** (Lee, 2009). This econometric technique trims the upper or lower tails of the outcome distribution in the less-attrited group, providing a mathematically rigorous worst-case and best-case scenario for the true Treatment Effect.

Chapter 10

The Pre-Registration Dilemma: Unregistered Outcomes

The advent of the credibility revolution led to the widespread adoption of Pre-Analysis Plans (PAPs) and platforms like the AEA RCT Registry. By publicly declaring primary endpoints and econometric specifications prior to data realization, researchers bind their hands, preventing p-hacking and Hypothesizing After the Results are Known (HARKing).

However, field experiments in applied microeconomics often yield unanticipated behavioral responses. What happens when a researcher discovers that the pre-registered primary outcome is flawed, while an unregistered exploratory variable captures a profound economic mechanism?

10.1 The “Moderation” Paradigm and the Populated PAP

As articulated by Banerjee, Duflo, et al. (2020), pre-registration should not act as an intellectual straitjacket that suppresses scientific discovery. To balance rigor and discovery, researchers must employ a **Dual-Track Strategy** utilizing a *Populated PAP*.

A **Populated PAP** is a standalone supplementary document that mechanically executes every empirical test exactly as specified in the original pre-registration, regardless of statistical significance. By providing this, the researcher honors their ex-ante commitments, eliminating concerns of the “file drawer” problem.

This liberates the main manuscript. The introduction, abstract, and primary results sections can safely and ethically lead with the *unregistered outcome*, prioritizing theoretical relevance over chronological compliance, provided the deviation is transparently labeled and justified.

10.2 Taxonomy of Scientific Justifications

When elevating an unregistered outcome to primary status in an academic journal, the researcher must include a dedicated “Deviation from Pre-Analysis Plan” section. Valid justifications fall into three typologies:

1. **Measurement Superiority:** The registered outcome suffered from severe measurement error, right-censoring, or differential non-response (e.g., self-reported income failed, but digital engagement metrics proved highly precise).
2. **Emergence of Unanticipated Mechanisms:** The intervention triggered behavioral responses (e.g., present-bias rejection of empathetic AI) that defy the original neoclassical model, rendering the pre-registered outcome theoretically irrelevant.
3. **Exogenous Shocks:** External events (e.g., a pandemic or macroeconomic crisis) fundamentally altered the context, shifting the policy relevance to a different set of variables.

10.3 The Econometric Imperative: Multiple Hypothesis Testing

Leading a paper with an unregistered primary outcome introduces profound statistical complications. Searching a dataset for significant outcomes inflates the **Family-Wise Error Rate (FWER)**. If a researcher tests 20 outcomes, the probability of finding at least one spurious result exceeds 64%.

To neutralize accusations of p-hacking, researchers must apply stringent algorithmic adjustments:

- **False Discovery Rate (FDR):** The Benjamini-Hochberg step-up procedure limits the expected proportion of rejected null hypotheses that are false positives, suitable for exploratory mechanisms.
- **The Romano-Wolf Bootstrap Procedure:** The gold standard for modern experimental economics. It uses resampling to account for the empirical dependence structure between highly correlated outcome variables.

Table 10.1: Reporting Unregistered Outcomes with MHT Adjustments

Exploratory Variable	Point Estimate (SE)	Naive p-value	Romano-Wolf Adjusted p-value
Unregistered Outcome Y_2	0.452 (0.112)	0.012**	0.038**
Unregistered Outcome Y_3	0.104 (0.098)	0.089*	0.214

Notes: Outcomes were not pre-registered. To correct for post-hoc selection bias and the expanded hypothesis space, p-values are adjusted using the Romano-Wolf step-down procedure across $m = 12$ observed outcomes.

If an unregistered outcome remains highly significant ($p < 0.05$) after a Romano-Wolf correction, it signals immense econometric rigor, proving the discovery is a robust economic mechanism rather than a statistical artifact.

Part IV

The Macro and Institutional Frontier

Chapter 11

Institutions, History, and Scale

11.1 The Political Economy of AI (Acemoglu)

The impact of AI on education is not pre-ordained by the algorithm; it is mediated by institutions. Does the AI tool *augment* teacher capabilities (an inclusive, labor-complementing path), or does it *automate* the pedagogical process to cut state budgets (an extractive, labor-replacing path)?

11.2 Historical Persistence (Dell)

AI interventions do not occur on a blank slate. By utilizing Machine Learning (Computer Vision and NLP) to digitize historical archives, we can trace how centuries-old state capacity constraints dictate the "last-mile" success of modern digital RCTs.

Chapter 12

General Equilibrium and Long-Term Mobility

12.1 Administrative RCTs and the Opportunity Atlas (Chetty)

To evaluate Becker's ultimate return on human capital, short-term test scores are insufficient. RCTs must be linked to administrative tax records. AI operates as a "digital neighborhood effect," capable of transcending zip codes and driving intergenerational geographic convergence.

12.2 Macroeconomic Spillovers (Blanchard)

Scaling AI education is a massive aggregate supply shock. If successful, it shifts the Long-Run Aggregate Supply (LRAS), compresses the skill premium through wage adjustments, and impacts the natural rate of unemployment (NAIRU). Furthermore, the initial capital expenditure of AI must be justified by the long-term human capital fiscal multiplier.

Conclusion

The experimental approach has transformed applied microeconomics. The transition from raw data to reliable causal knowledge relies entirely on the successful elimination of selection bias. By integrating Becker's human capital model with clean identification strategies (LATE), structural behavioral modeling (MTE), and macroeconomic realities, this book provides the definitive framework for policymakers navigating the digital revolution.